

Communications in Virtual Environment Improve Interpersonal Impression

Yuki Kato ¹Toyohashi University of
TechnologyMaki Sugimoto ²

Keio University

Masahiro Inami ³

The University of Tokyo

Michiteru Kitazaki ⁴Toyohashi University of
Technology

ABSTRACT

Pseudo physical touch is used for communications in virtual environments such as VRChat. We aimed to test if the pseudo-touch communication affects social impression in a virtual environment. Nineteen participants performed the controlled experiment with a partner who was an experimenter with three types of communications: no touch, pseudo touch, and actual touch. Subjective ratings of attractiveness and the communication easiness with the partner increased in all conditions, suggesting that the communication in virtual environments improves interpersonal attraction and communicability either with or without physical or pseudo touch.

Keywords: Pseudo touch, communication, interpersonal impression

Index Terms: I.3.7 [Computer graphics]: Three-Dimensional Graphics and Realism—Virtual Reality; H.1.2 [Models and Principles]: User/Machine Systems—human factors

1 INTRODUCTION

Communications in cyberspace is getting popular, during COVID-19 pandemic in particular. VRChat that is a social virtual reality (VR) networking application had 7,600 concurrent users in July 2019, then the users increased to 16,000 in February 2020 [1]. However, due to its rapid growing of the VRChat society, users engaging in disrespectful or harmful behaviors appeared [2]. To maintain VR social networking services to be fun and safe, it is important to study what factors affect interpersonal social perception and behavior of users.

Social touch or interpersonal touch improves non-verbal communications, particularly in emotional relationship [3]. Social touch enhances prosocial behavior [4] and interpersonal impression [5]. In the human-computer interaction research field, various haptic communication devices and systems have been developed [2, 6-7]. In VRChat, users sometime touch each other with avatar bodies, and such behaviors could improve interpersonal impression and communication [8]. However, the touch in VRChat lacks actual tactile or force sensation and is based on only visual sensation, similarly to pseudo haptics [9].

¹ kato@real.cs.tut.ac.jp

² sugimoto@ics.keio.ac.jp

³ inami@inami.info

⁴ mich@tut.jp

We aimed to test whether actual-touch, pseudo-touch, and non-haptic communications affect social impression in a virtual environment.

2 METHODS

2.1 Participants

Nineteen university students (All males, mean age 21, SD 0.258) participated in the experiment. They were paired with a partner who was an experimenter, but they did not know it; we informed that the partner was one of volunteers like themselves. All participants were naive to the purpose of the experiments, and had normal or corrected-to-normal vision with normal exercise capacity. They were provided informed consent before the experiment. The methods of the experiment were approved by the Ethical Committee for Human-Subject Research at the Toyohashi University of Technology.

2.2 Apparatus

Two participants' bodies were monitored by a motion capture system (Optitrack Flex 3, 12 cameras, 120fps) with a computer (Intel Core i7-6700K, NVIDIA Quadro P600), and the data were streamed to other computers for presenting avatars in a virtual environment on two head-mounted displays (HMD). We used two computers (Intel Core i7 8700, NVIDIA GeForce GTX 1070Ti) and Unity (2018.4.8f1) to create and present a virtual environment to the two HMD (HTC VIVE Pro Eye, 1440 [width] × 1600 [height] pixels, refresh rate 90Hz).

2.3 Stimuli and Conditions

We set three communication conditions (Actual touch, Pseudo touch, and No touch), and assigned one avatar for each condition (Figure 1). Correspondence of avatars and the conditions were randomized across participants. The experiments were performed in three different days (a condition per day) in a random order.



Figure 1: Avatars of partner and participants.

In the actual touch condition, the participant and the partner checked the head each other in the real and the virtual environments.

In the pseudo touch condition, the avatars of the participant and the partner touched the head each other in the virtual environment, but they did not actually touch in the real environment by distancing them (Figure 2). In the no touch condition, they waved the hand each other.

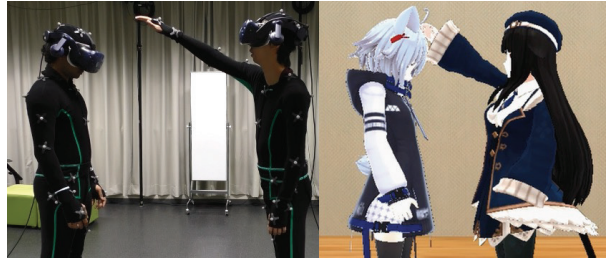


Figure 2: Schematic of the pseudo-touch condition. Participants did not touch in the real environment (Left), but visually touched in the virtual environment (Right).

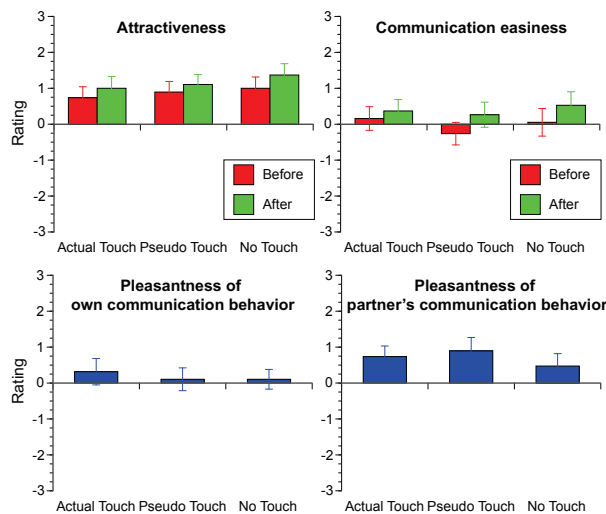


Figure 3: Results of Experiment. Bar plots indicate means +/- SEM.

2.4 Procedure

Participants were asked to answer the questionnaire on attractiveness and communication easiness before and after the communications in 7 level Likert scale (-3: I do not agree at all, +3: I agree very much). Questionnaire items were as follows:

- The partner is attractive.
- I can communicate with the partner easily.

Additional two items were asked only after the communications:

- I felt pleasant when I performed the communication behavior to the partner.
- I felt pleasant when I was performed the communication behavior by the partner.

3 RESULTS

Two-way repeated measures ANOVAs (before/after communication x 3 types of communication) were conducted for the attractiveness and the communication easiness scores. We found a main effect of the before-after condition for the attractiveness ($F(1,18)=5.98$, $p=0.025$, $\eta_p^2=0.250$), but did not find a main effect of communication type condition ($F(2,36)=1.61$, $p=0.215$, $\eta_p^2=0.082$) or the interaction ($F(2,36)=0.28$, $p=0.757$, $\eta_p^2=0.015$). We found a main effect of the before-after condition the communication easiness ($F(1,18)=7.70$, $p=0.0125$, $\eta_p^2=0.300$) but did not find a main effect of communication type condition

($F(2,36)=0.83$, $p=0.443$, $\eta_p^2=0.044$) or the interaction ($F(2,36)=0.85$, $p=0.434$, $\eta_p^2=0.045$). The results showed that the attractiveness and the communication easiness were improved after the communication in the virtual environment irrespective of communication types (Figure 3 Top). One-way repeated measures ANOVAs for the pleasantness of own and partner's communications did not indicate any main effects (Own: $F(2,36)=0.44$, $p=0.646$, $\eta_p^2=0.024$, Partner's $F(2,36)=0.73$, $p=0.488$, $\eta_p^2=0.039$).

4 DISCUSSION

The results showed that subjective ratings of attractiveness and the communication easiness with the partner were increased by three types of communication in the virtual environment: actual touch, pseudo touch, and waving the hand. There was not significant difference between effects of communication types on interpersonal impression. These results suggest that the communication in virtual environments increases interpersonal attraction and communicability either with or without actual or pseudo touch. However, the avatars' appearance might limit the effect of pseudo touch. The avatars in the experiment could be too attractive for participants to differentiate the effects of actual touch, pseudo touch and waving hands; like a ceiling effect. We should use moderately attractive avatars in a future study.

We speculate that communications such as physical touch and waving hands could be effective for prevent users' harmful and disrespectful behaviors, based on our results. During COVID-19 pandemic, pseudo-haptic communication could be a method for safe and valuable communications.

ACKNOWLEDGMENTS

This research was partly supported by JST ERATO Grant Number JPMJER1701 (Inami JIZAI Body Project), and KAKENHI 20H04489.

REFERENCES

- [1] Road to VR. Social VR App 'VRChat' is Seeing Record Usage Amidst the Pandemic, <https://www.roadtovr.com/vrchat-record-users-coronavirus/>, 2020. (retrieved 23rd December 2020).
- [2] VRChat. An open letter to our community, <https://medium.com/@vrchat/an-open-letter-to-our-community-1b7aa5d9026f>, 2018. retrieved 23rd December 2020).
- [3] A. Gallace, & C. Spence. The science of interpersonal touch: an overview. *Neuroscience & Biobehavioral Reviews*, 34(2), 246-259. 2010.
- [4] J. K. Burgoon, J. B. Walther, & E. J. Baesler Interpretations, evaluations, and consequences of interpersonal touch. *Human Communication Research*, 19(2), 237-263. 1992.
- [5] N. Guéguen, & J. Fischer-Lokou. Another Evaluation of Touch and Helping Behavior. *Psychological Reports*, 92(1), 62-64. 2003.
- [6] D. Tsetserukou. HaptiHug: A novel haptic display for communication of hug over a distance. In *International Conference on Human Haptic Sensing and Touch Enabled Computer Applications* (pp. 340-347). Springer, Berlin, Heidelberg. 2010
- [7] N. Takahashi, Y. Kuniyasu, M. Sato, S. Fukushima, M. Furukawa, Y. Hashimoto, H. Kajimoto, Kiss Interface for Intimate Communications, *Journal of Human Interface Society*, Vol.13, No.4, pp.53-62, 2011(in Japanese)
- [8] K. Nagamachi, Y. Kato, M. Sugimoto, M. Inami, & M. Kitazaki. Pseudo Physical Contact and Communication in VRChat: A Study with Survey Method in Japanese Users, *ICAT-EGVE 2020*, DOI: 10.2312/egve.20201275, 2020.
- [9] A. Lécuyer, J. M. Burkhardt, & L. Etienne. Feeling bumps and holes without a haptic interface: the perception of pseudo-haptic textures. In *Proceedings of the SIGCHI conference on Human factors in computing systems*, 239-246, 2004.